RE-MESH

25-26J-011

SLIIT FACULTY OF COMPUTING

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Introduction

The Scenario

- Sri Lanka faces devastating floods every year
- Homes, farmlands, and livelihoods destroyed
- Thousands of families displaced annually
- Communities lack access to real-time flood warnings

The Problem

- Late warnings leave families little time to evacuate
- Water levels rise faster than communication spreads
- Internet dependency fails during emergencies
- Manual reporting systems collapse under pressure

Mesh Node Hardware & Deployment

Specialization: Computer Systems and Network Engineering

Presenter

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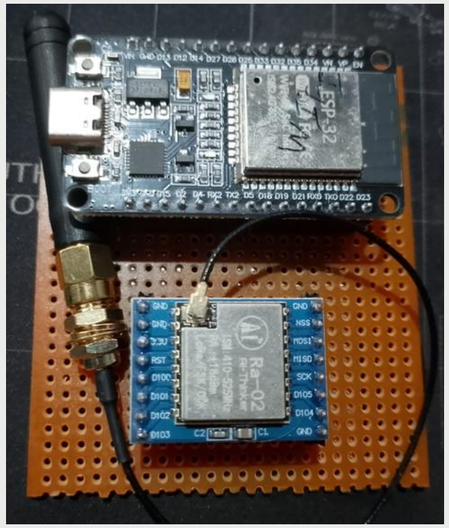
Hardware Prototype & Problem Fit

Prototype Overview

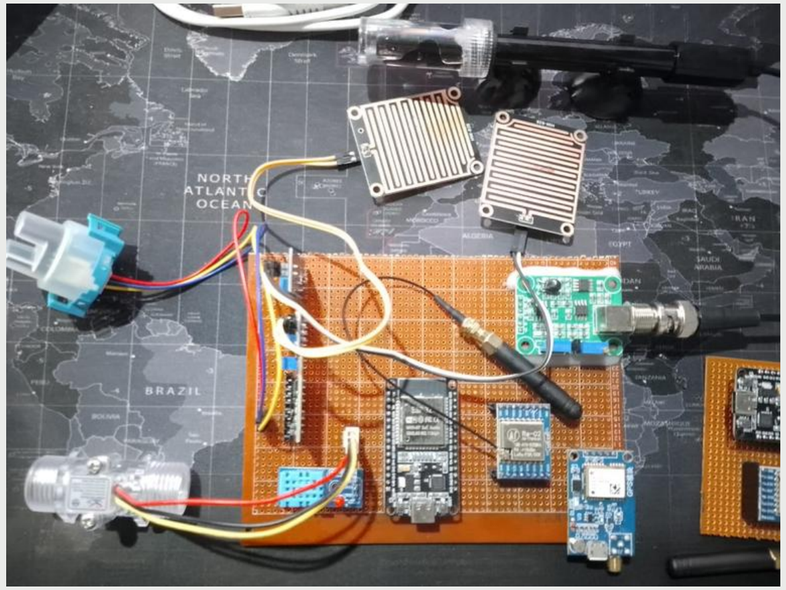
- 3x ESP32 + LoRa nodes
  - Sensor Node: water level & environmental data
  - Gateway Node: LoRa → Wi-Fi, local web visualization
  - End Device: LoRa client, future analytics integration



How It Addresses the Sub-Problem

- Enables early flood indication near dams
- Operates during internet & power failures
- Supports local decision-making via Wi-Fi access

Hardware Prototype & Problem Fit Cont.



User Requirements Addressed

- Real-time monitoring
- Long-range, low-power communication
- Offline accessibility

Design Contribution

- Edge-based monitoring using LoRa (LPWAN)
- Decentralized gateway architecture
- Designed for future solar power & rugged enclosure

LPWAN technologies like LoRa are widely used in flood monitoring due to low power consumption and long-range reliability
(e.g., Centenaro et al., 2016; Mekki et al., 2019)

Sustainability & Research Alignment

Target Users

- Disaster Management Centre
- Irrigation & Dam Authorities
- Flood-prone communities

What Makes ReMesh Unique

- Functions without continuous internet
- Local web access + LoRa resilience
- Community-first design over cloud dependency

Sustainability & Research Alignment Cont.

Commercialization & Sustainability

- Government & NGO deployments
- Hardware + monitoring service model
- Expandable to regional flood-prone areas

Research Relevance

- Aligns with IoT-based flood monitoring literature
- Supports edge intelligence over cloud-only systems
- Designed for scalable disaster response networks

Edge-based disaster monitoring improves response time and system reliability (Satyanarayanan, 2017; Zhang et al., 2022)

Flood Prediction and End-User Dashboard

Specialization: Computer Systems and Network Engineering

Presenter

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How the Solution Addresses the Sub-Problem

Problem Identified

- During floods in Sri Lanka, communities lack **real-time predictive warnings** and **accessible communication interfaces**, leading to delayed evacuations and increased casualties

Solution:

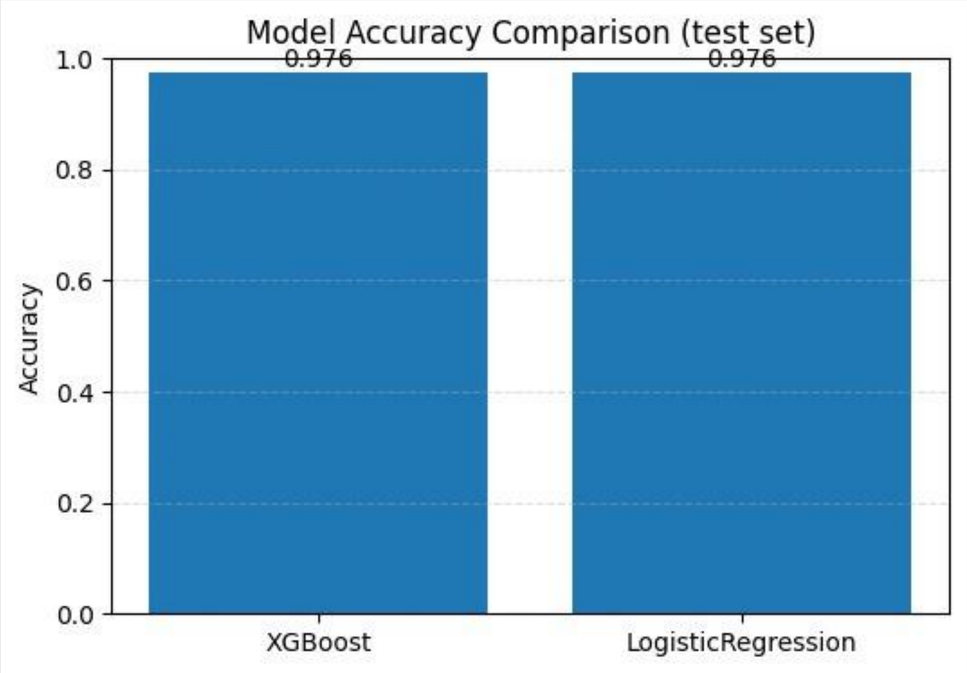
- ML-Based Prediction -XGBoost & Logistic Regression models analyze 8 environmental parameters with hyperparameter tuning
- End-User Dashboard - Intuitive interface displays real-time flood risk levels, sensor data visualization, and emergency messaging capabilities for affected communities
- Real-Time Integration
- Direct integration with ESP32 hardware nodes via LoRa, processing sensor data every 5 minutes for immediate risk assessment

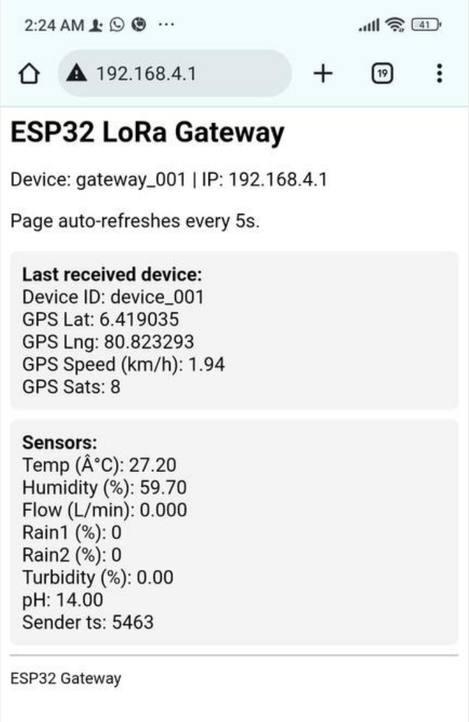
User Requirements

- Flood Risk Prediction
- Offline Capability
- Real-Time Monitoring
- Low Resource Usage

Design Excellence

- Ensemble ML Approach
- Hyperparameter Tuning
- Progressive Web App (PWA) Architecture





Prototype Status

Completed Features

- ML model training with hyperparameter tuning
- Real-time data visualization dashboard
- Dashboard Integration with ESP32 test nodes

In Progress

- Full integration with physical sensor nodes for real-time data streaming
- Dashboard UI/UX enhancement for better visualization and user experience
- Displaying emergency alerts in dashboard.
- Rainfall prediction integration using Weather API data

Re-Mesh Routing & Communication Module

Specialization: Computer Systems and Network Engineering

Presenter

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Proven Gap & Creative Solution

Sub-Problem Addressed

Communication failure during floods due to:

- Cellular and internet infrastructure breakdown
- Power outages
- Fixed network dependency

Need for reliable node-to-central communication in disaster zones

Proposed Solution

- Design of a LoRa-based Re-Mesh routing mechanism
- Nodes broadcast data using LoRa when traditional networks fail
- Dynamic routing between nodes to reach the gateway
- Automatic re-routing when intermediate nodes fail

Proven Gap & Creative Solution

How the Solution Addresses the Sub-Problem

- Operates independently of ISP or cellular networks
- Maintains communication even with partial node failures
- Enables continuous data flow from disaster areas to the central dashboard
- Supports rapid deployment with low power and low cost

User Requirements & Design Excellence

User Requirements Addressed

- Reliable communication during flood disasters
- Long-range, low-power data transmission
- Identification of active and inactive nodes
- Real-time environmental data access
- Local monitoring when internet is unavailable
- Ability to scale with additional nodes

Design Excellence / Contribution

- Disaster-aware self-healing LoRa mesh routing
- Broadcast-based, ID-centric communication
- Failure detection and re-routing
- Lightweight gateway design using ESP32
- Routing designed specifically for constrained IoT devices

Prototype Status & Progress

Prototype Status

- Functional LoRa communication between nodes and gateway
- Gateway receives, decodes, and displays sensor data
- Heartbeat mechanism implemented for node monitoring

Completed Features

- Unique node ID–based packet transmission
- Broadcast-based routing logic
- JSON-based payload structure
- Sensor data forwarding to local server

Features In Progress

- Node failure detection through the ID binding
- Lightweight packet-level encryption for ESP32

Centralized Monitoring & Management System

Specialization: Computer Systems and Network Engineering

Presenter

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Central Monitoring & Management System for RE-MESH

Problem Identified:

Lack of real-time centralized monitoring and management system for disaster communication mesh networks, making it difficult to track node status, coordinate emergency response, and manage devices during critical situations.

Solution Implemented:

Completed Features ✅

- Device status monitoring interface
- Data gathering & visualization
- Location tracking
- Device registration system
- Analytical dashboard views

Pending Development ⏳

- Real-time IoT device integration
- Voice/text emergency alerts
- Real-time data communication
- Enhanced UI/UX refinement

User Requirements Addressed:

Emergency Response Teams

✓ Real-time node health monitoring
✓ Geographic device positioning
✓ Centralized command interface
✓ Quick device registration

System Administrators

✓ Centralized management
✓ Performance analytics
✓ Data logging & reporting
✓ Offline operation capability

Design Excellence & Contribution:

Modular Architecture

- MVC separates data, logic, and presentation, enhancing scalability, maintainability, and reliability for system.

RESTful API Design

- Standardized JSON communication protocol enables seamless IoT device integration and third-party interoperability

Responsive Interface

- Intuitive dashboard accessible on various devices for field deployment and command center operations

Commercialization & Sustainability

Market Positioning and Business Model

Customer Persona –

- The system targets disaster management authorities, emergency responders, smart city administrators, and IoT solution providers who require reliable real-time monitoring platforms.

Market Size & Opportunity –

- The increasing adoption of disaster monitoring, smart emergency response, and IoT-based situational awareness systems creates strong market potential.

What Makes the Solution Unique –

- The solution offers a low-cost, IoT-ready centralized monitoring platform with real-time and historical analytics and future emergency alert integration.

Cost Recovery & Sustainability –

- Sustainability is achieved through software licensing, subscription-based analytics, custom deployment services, and an open-core revenue model.

THANK YOU